

Ime in priimek:

MITJA ŠEVERKAR



dosežene točke	možne točke	odstotki	ocena
35	38	92	5

ČAS PISANJA: 45 minut

1. Polinom $p(x) = 2x^4 + ax^3 + x^2 + 5$ da pri deljenju s polinomom $x + 2$ za 30 manjši ostanek kot pri deljenju s polinomom $x - 3$. Izračunaj vrednost realnega števila a .

[5t] 5

$$\begin{array}{r|rrrrr} & 2 & a & 1 & 0 & 5 \\ -2 & & -4 & -2a+8 & 4a-18 & -8a+36 \\ \hline & 2 & a-4 & -2a+9 & 4a-18 & -8a+41 \end{array}$$

$$-8a+41+30=27a+176$$

$$-35a=105$$

$$a=-3$$

$$\begin{array}{r|rrrrr} & 2 & a & 1 & 0 & 5 \\ 3 & & 6 & 3a+18 & 9a+57 & 27a+171 \\ \hline & 2 & a+6 & 3a+19 & 9a+57 & 27a+176 \end{array}$$

2. Dan je polinom $p(x) = 8x^2 - 4x + 5$. Določi predpis polinoma q (v poenostavljeni obliki), za katerega velja $p(x) = q(2x - 1)$.

[4t] 4

$$8x^2 - 4x + 5 = a(x+b)(2x-1) = a(2x^2 - x + 2bx - b) + c$$

$$\Rightarrow a=4$$

$$8x^2 - 4x + 8bx - 4b = 8x^2 - 4x + 5 \Rightarrow b=0, c=5$$

$$q(x) = 2x^2 + 2x + 5 = 2(x^2 + x + \frac{5}{2})$$

$$q(x) = 4x^2 + 4x + 5$$

$$q(x) = 4x^2 + 4x + 5$$

$$a=2$$

$$-8x + 2bx = -4x$$

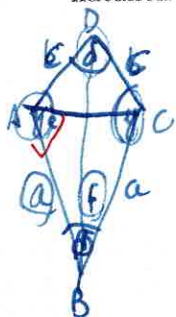
$$2b=4$$

$$b=2$$

$$c=5$$

$$\begin{aligned} q(x) &= ax^2 + bx + c = \\ &= a(2x-1)^2 + b(2x-1) + c = \\ &= a(4x^2 - 4x + 1) + 2bx - b + c = \\ &= 4ax^2 - 4ax + a + 2bx - b + c = \\ &= 8x^2 - 8x + 2bx + 2 - b + c \end{aligned}$$

3. Dan je deltoid $ABCD$ s podatki $a = |AB| = 8 \text{ cm}$, $f = 11 \text{ cm}$ in $\gamma = 110^\circ$. Izračunaj vse notranje kote (na minuto natančno) in obseg deltoida (na eno decimalno mesto natančno). [7t] **7**



$$\frac{f}{\sin \gamma} = \frac{a}{\sin d_1}$$

$$\sin d_1 = \frac{a \cdot \sin \gamma}{f}$$

$$d_1 = 43,1^\circ$$

$$d = 2d_1 = 86,2^\circ = 86^\circ 13'$$

$$\beta = 360^\circ - 2\gamma - d = 53^\circ 47'$$

$$\frac{f}{\sin \gamma} = \frac{c}{\sin \frac{\beta}{2}}$$

$$c = \frac{f \cdot \sin \frac{\beta}{2}}{\sin \gamma} = 5,29 \text{ cm}$$

$$o = 2(a+c) = 26,6 \text{ cm}$$

4. Pokončna, 8 cm visoka, tristrana piramida s prostornino $32\sqrt{3} \text{ cm}^3$ ima osnovna robova dolga 6 cm in 8 cm. Osnovna ploskev je ostrokotni trikotnik. Izračunaj kot med stranskim robom in osnovno ploskvijo. Kot naj bo na desetinko stopinje natančen. [9t] **6**



$$v = 8 \text{ cm}$$

$$V = 32\sqrt{3} \text{ cm}^3 = \frac{O \cdot v}{3}$$

$$96\sqrt{3} = O \cdot v \Rightarrow O = 12\sqrt{3}$$

$$a = 6 \text{ cm}$$

$$b = 8 \text{ cm}$$

$$s = \frac{a+b+c}{2}$$

$$= \frac{14+c}{2}$$

$$= 7 + \frac{c}{2}$$

$$O = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{\left(7 + \frac{c}{2}\right)\left(1 + \frac{c}{2}\right)\left(-1 + \frac{c}{2}\right)}$$

$$= \sqrt{\left(\frac{c^2}{4} - 1\right)\left(7 + \frac{c}{2}\right)7}$$

$$R = \frac{abc}{4s}$$

$$R = \frac{6 \cdot 8 \cdot c}{4 \cdot 12\sqrt{3}} = 2\sqrt{3} c$$

$$\frac{v}{\sin \alpha} = 2R$$

$$\sin \alpha = \frac{v}{2R} = \frac{4}{R} = \frac{2\sqrt{3}}{3} c$$

$$432 = \left(\frac{c^2}{4} - 1\right)\left(\frac{c}{2} + 7\right)$$

$$432 = \left(\frac{c^2}{4} - 1\right)\left(\frac{c}{2} + 7\right)$$

$$432 = \frac{c^3}{8} + \frac{7c^2}{4} - \frac{c}{2} - 7$$

$$3456 = 7(c^3 + 14c^2 - 4c - 56)$$

$$7c^3 + 98c^2 - 28c - 392 = 3456$$

$$c = 5,423832957$$

5. Krogla in pokončni valj imata enaki površini. Polmer krogle je dolg trikrat toliko kot polmer osnovne ploskve valja, prostornina krogle pa je za $1216\pi \text{ cm}^3$ večja od prostornine valja. Izračunaj polmer osnovne ploskve valja. [7t] 7



$$P_o = 4\pi R^2 \quad V_o = \frac{4\pi R^3}{3}$$

$$P_v = 2\pi r^2 + 2\pi r \cdot v = 2\pi r(r+v) \quad V_v = \pi r^2 v$$

$$R = 3r$$

$$4\pi R^2 = 2\pi r(r+v)$$

$$36\pi r^2 = 2\pi r(r+v)$$

$$36\pi r - 2\pi r = 2\pi v$$

$$34\pi r = 2\pi v$$

$$v = 17r$$

$$\pi r^2 v + 1216\pi \text{ cm}^3 = \frac{4\pi R^3}{3}$$

$$17\pi r^3 + 1216\pi \text{ cm}^3 = \frac{4\pi (3r)^3}{3}$$

$$-19\pi r^3 = -1216\pi \text{ cm}^3$$

$$r = \sqrt[3]{\frac{1216 \text{ cm}^3}{19}} = 4 \text{ cm}$$

6. Lik na sliki zavrtimo za 360° okrog osi (glej sliko). Izračunaj prostornino in površino vrtenine. Rezultata naj bosta točna. [6t] 6

$$V_n = \frac{\pi x^2 \cdot v}{3} = 324\pi$$

$$V_z = \frac{\pi R^2 \cdot v}{3} = 1024\pi$$

$$V = 700\pi$$

$$P_{I_n} = \frac{2\pi x \cdot 15}{2} = 135\pi$$

$$P_{I_z} = \pi R \cdot 20 = 320\pi$$

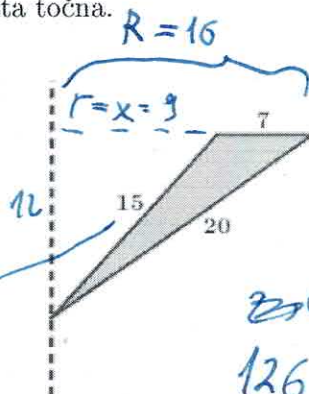
$$O_n = \pi x^2 = 81\pi$$

$$O_z = \pi R^2 = 256\pi$$

$$O = 175\pi \quad O_z - O_n = 175\pi$$

$$P_I = P_{I_n} + P_{I_z} = 455\pi$$

$$P = O + P_I = 630\pi$$



$$v^2 = 15^2 - x^2$$

$$v^2 = 20^2 - (7+x)^2$$

$$400 - 49 - 14x - x^2 = 225 - x^2$$

$$126 = 14x$$

$$x = 9$$

$$v = \sqrt{15^2 - 9^2} = 12$$